# LegalEase

AI-Powered Legal Document Generator

**PHASE 4 • PROJECT DEVELOPMENT**

Development • Integration • Validation • Document Generation

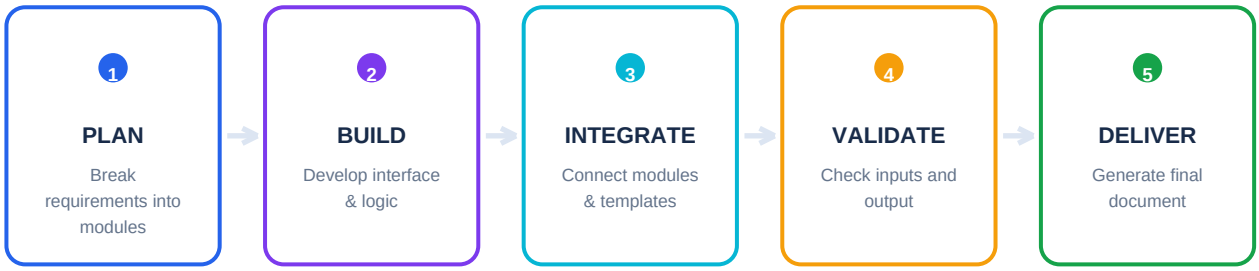
College Project Documentation

**Project Development** is the implementation stage where the requirements and design are converted into functional software. For LegalEase, the development process focuses on collecting user inputs, applying legal-document templates and logic, generating structured content, validating the result, and producing a readable document for the user.

## Development Objectives

- Build a simple and user-friendly interface for entering legal document details.
- Organize document templates and reusable fields so documents can be generated consistently.
- Connect the input, legal logic/AI assistance, validation, and output modules.
- Generate properly structured legal documents with headings, clauses, dates, names and other required fields.
- Keep the generated output readable, editable/previewable and suitable for download.

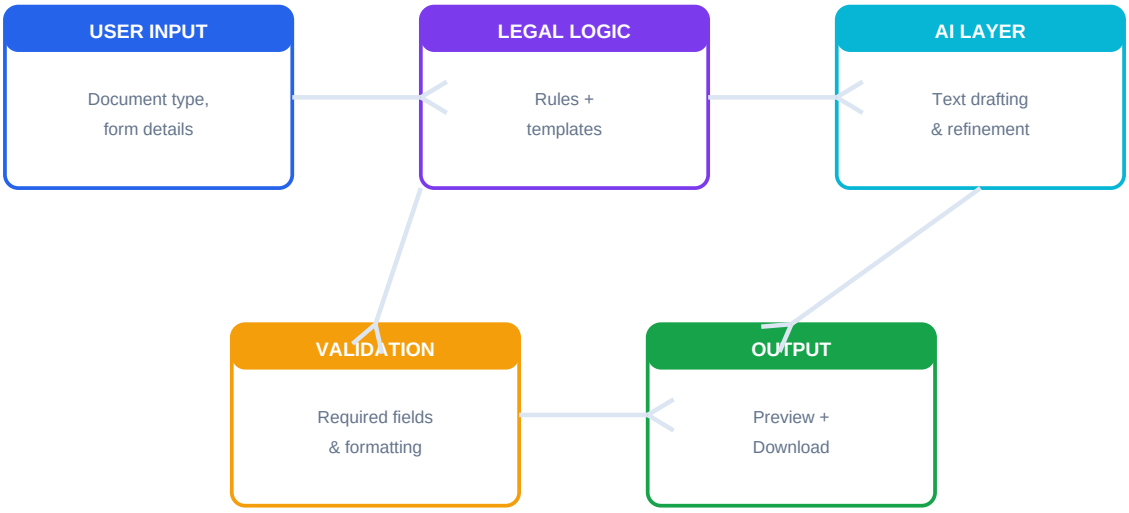
## Development Cycle




**Key idea:** LegalEase follows a modular approach, so each major function can be developed and improved without changing the whole application.

# 2. System Architecture & Modules

How the major development components work together




## Main Development Modules

| Module | Development Responsibility | Output |
|---|---|---|
| User Interface | Accepts document type, personal details, dates and other required information. | Validated form data |
| Template / Legal Logic | Maps user inputs to document structure, clauses and placeholders. | Structured legal content |
| AI Assistance | Supports drafting/refinement of natural-language content while preserving the required structure. | Improved text |
| Validation | Checks missing fields, basic formatting and document completeness. | Validated content |
| Document Generator | Combines content into a formatted document for preview/download. | Final legal document |

# 3. Implementation Workflow

Step-by-step flow used during LegalEase development

## User-to-Document Flow

## Example of Development Logic

```
1. Receive document type and user details
2. Check required fields
3. Load the matching document template
4. Insert validated values into placeholders
5. Generate/refine content where required
6. Format headings, clauses and signatures
7. Create preview and downloadable document
```

## Development Practices

- **Reusable components:** common fields and formatting are designed for reuse across different document types.
- **Input validation:** incomplete or incorrectly formatted essential fields are identified before document generation.
- **Separation of modules:** interface, document logic and output generation are kept conceptually separate.
- **Readable output:** headings, paragraphs, clauses and signature areas are arranged in a clear order.

# 4. Development Validation & Testing

Checking whether each implemented component behaves as expected

During development, LegalEase is checked module-by-module and then as a complete workflow. The goal is to catch missing inputs, formatting problems and incorrect document assembly before the final document is delivered.

## Sample Development Test Cases

| Test | Input / Action | Expected Result |
|---|---|---|
| TC-01 | Open application and select document type | Relevant form/template is displayed |
| TC-02 | Submit with required field missing | Validation identifies the missing information |
| TC-03 | Enter valid details and generate | Structured document is produced |
| TC-04 | Review generated text and formatting | Content appears in the correct sections |
| TC-05 | Use preview/download option | Generated file is available to the user |

## Quality Checklist

| | |
|---|---|
| ✓ | **Input** — Required details are collected clearly. |
| ✓ | **Logic** — Selected document type maps to the intended structure. |
| ✓ | **Formatting** — Headings, clauses and fields follow a consistent layout. |
| ✓ | **Output** — Final document can be previewed/generated without broken sections. |

**Important:** LegalEase is a document-generation project; generated documents should be reviewed by an appropriate person for the specific legal situation before real-world use.

## 5. Phase 4 Outcome

What is achieved at the end of Project Development

At the end of Phase 4, the LegalEase concept is converted into an implementable software workflow. The major modules are connected so that a user can provide information, the system can process it through the designed logic, and a structured legal document can be generated as the final output.

| | |
|---|---|
| 01 | **Functional workflow**<br>User input → processing → validation → document generation |
| 02 | **Modular design**<br>Separate components make the project easier to maintain and extend |
| 03 | **Consistent output**<br>Templates and formatting help maintain a clear document structure |
| 04 | **Validation layer**<br>Essential fields and basic output quality are checked before delivery |
| 05 | **Ready for next phase**<br>The developed workflow can be demonstrated and documented for testing/evaluation |

### Phase 4 Summary

LegalEase development combines a user-friendly interface, reusable document structures, processing logic, validation, and formatted output. This phase establishes the working foundation required for the later testing, documentation and demonstration activities.

**LEGAL EASE**
**From user details to a structured legal document — developed as a simple, organized and reusable workflow.**